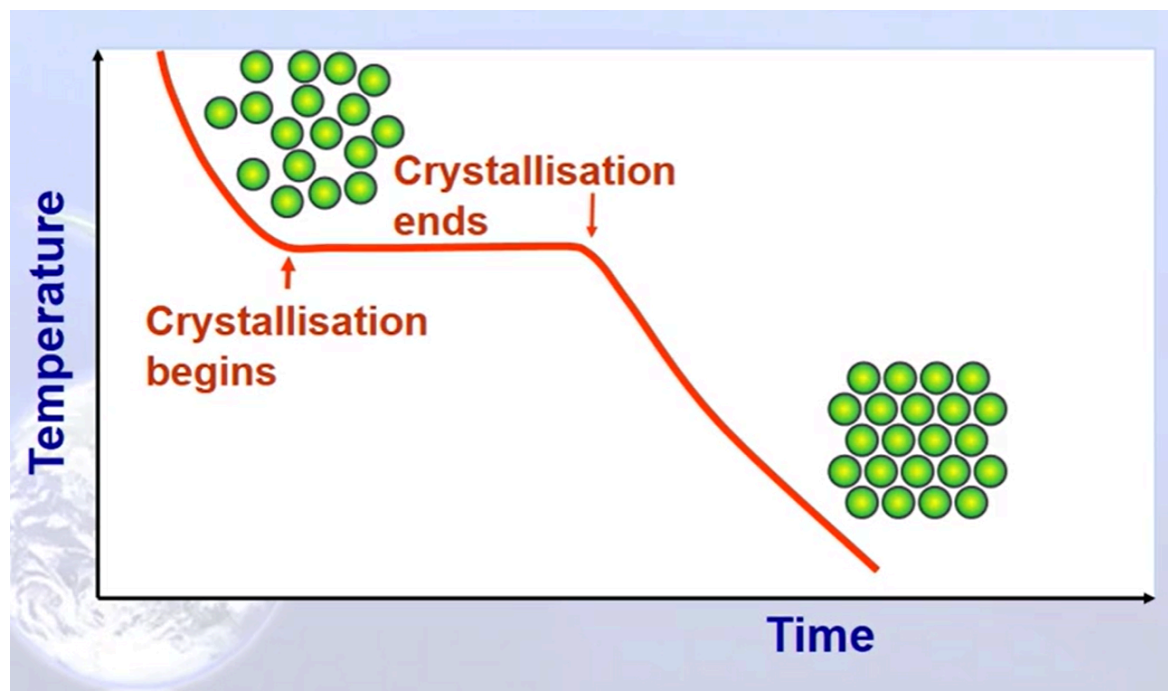


# Plain Carbon Steels

## Crystal Structure

Crystalline structure refers to an orderly, regular and geometric arrangement of atoms in the material in all 3 dimensions.

Most metals when solid have this crystalline structure, however when they are a liquid, the atoms are not arranged in a regular manner, due to the bonds being weak.



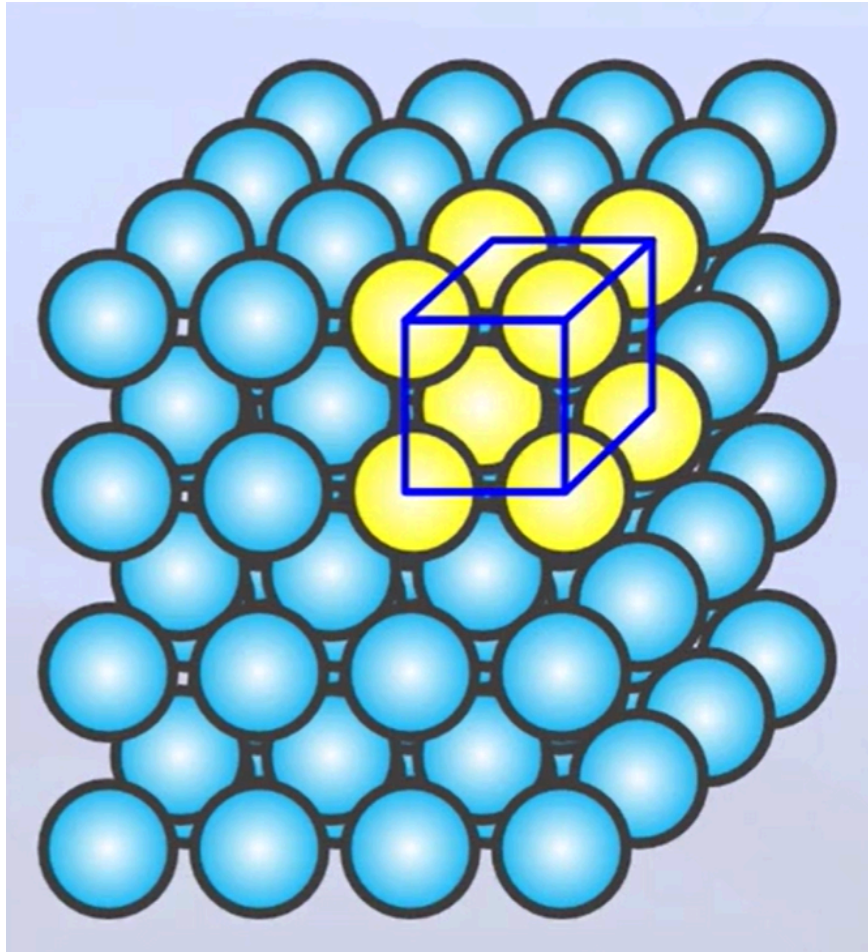
When metals cool from liquids to solids, the atoms rearrange themselves as temperature decreases, arranging back into their crystalline forms.

When the temp is at the melting point, crystallization occurs, which is when the atoms rearrange themselves.

During crystallisation, the atoms lose energy as the forces of attraction become stronger. The energy is lost as latent heat (recall physics  $Q=ml$ , total energy lost is latent heat of fusion; latent refers to change of state)

The end crystalline structure is also known as a lattice structure.

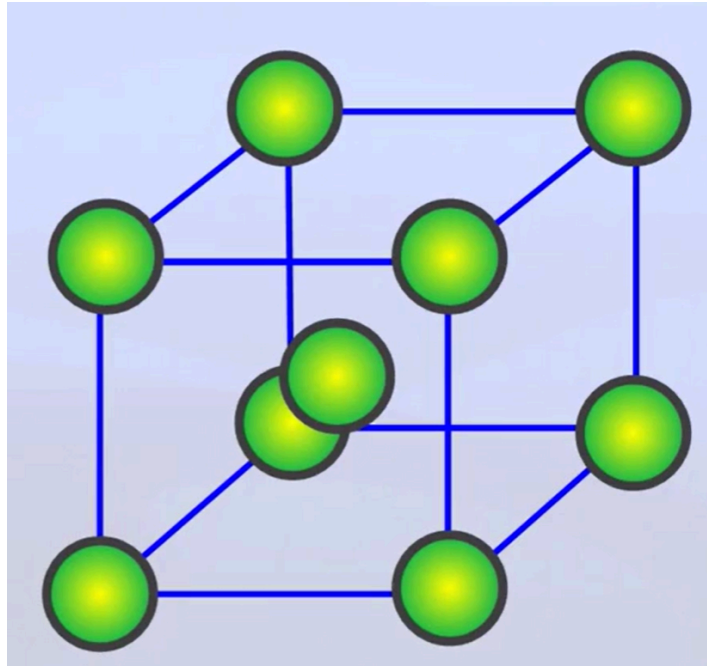
As these structures are symmetric and repeating throughout, we can simplify the lattice structure into the unit cell, the smallest group that is repeated throughout the structure.



There are some common crystalline and unit cell structure for most metals:

#### Body-Centered Cubic (BCC)

This structure consists of a cubic unit cell, with atoms at each corner and another atom at the center of the cube body (hence body-centered)

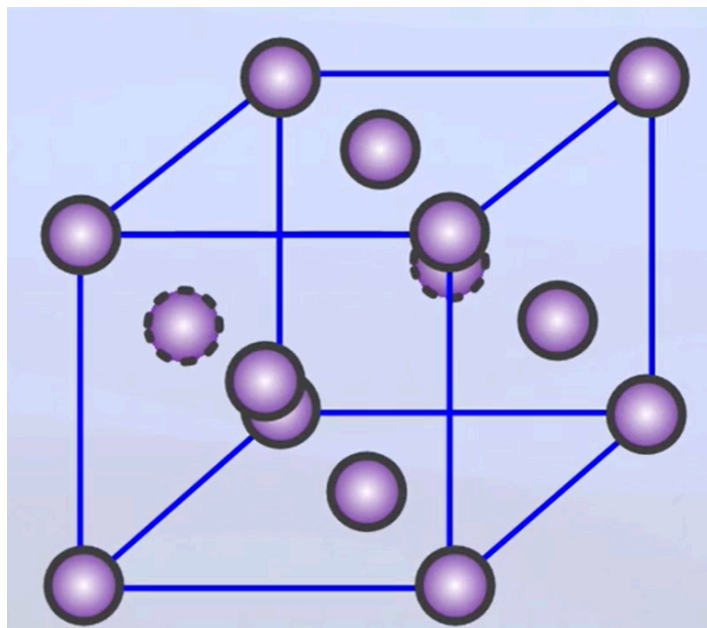


There are  $8+1=9$  atoms in this unit cell.

some metals that have this structure are chromium,  $\alpha$ -iron(ferrite), and tungsten. (C[u]rFeW)

#### Face-centered Cubic (FCC)

This structure is also a cubic cell, having an atom at each corner, however with extra atoms at the center of each face of the cube.



This structure consists of  $8+6 = 14$  atoms.

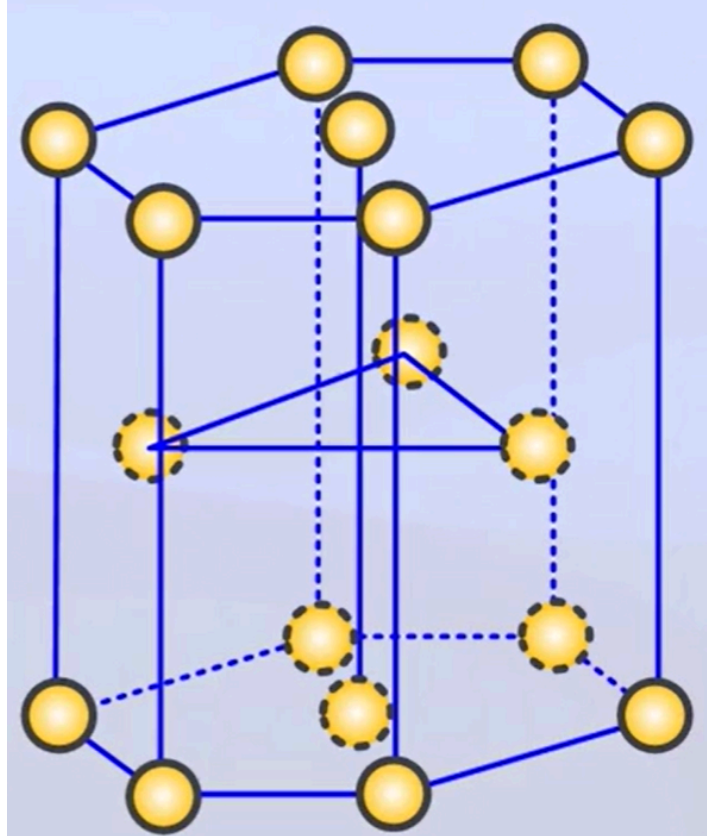
Metals than have this structure are:

Aluminium, copper, nickel, gold, silver (Al Cu Ni Au Si) and austenite

Al Cu Au Ni Si

Close packed Hexagonal (HCP or CPH)

This structure consists of a hexagonal prism unit cell, with 2 atoms at the poles/hexagonal faces, and 3 atoms arranged in a triangle in the middle of the cell, where the triangle is parallel to the hexagonal faces.



This structure consists of  $2(6)+2+3 = 17$  atoms.

Metals that have this structure are Mg Zn Ti(at low temp)

rmb the poles !!!!! please

Other than crystal structures, there are also grain structures, where crystal structures focus on the atoms of metals, while grain structures focus on a larger scale.

Solidification of crystals

When producing solid metals, it is most commonly produced via cooling pure molten metal.



When crystallization starts, small nuclei in the molten metal forms at various positions throughout the metal, with different orientations. These act as the center of crystal growth.

These crystals start to grow into dendrites (branch like arms), where they continue to grow, making the crystal(s) bigger.

▼ Why do dendrites form:

The surface of the crystal will be uneven and thus small imperfections on the surface of the crystal may/will jut out.

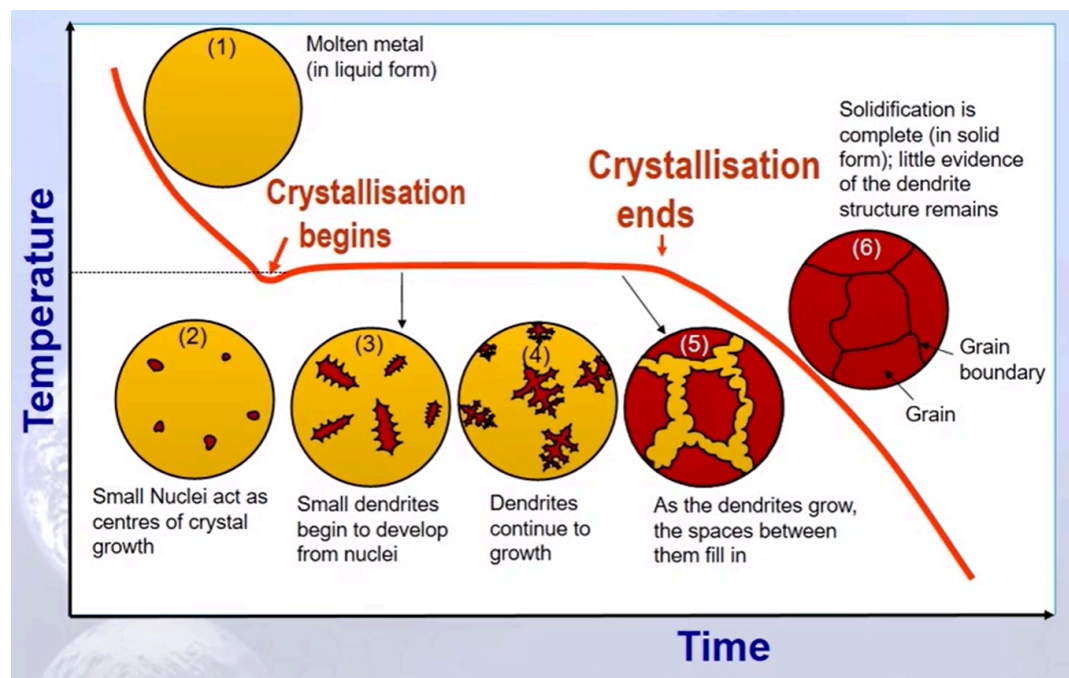
Those imperfections are now closer to the cooler edges of the mould, thus the protrusion is now cooler than the surface of the metal.

As the protrusion is cooler, it will continue to cool down faster and the protrusion will be longer. As the protrusion gets longer, more imperfections on the surface of the branch will also jut out, repeating the process

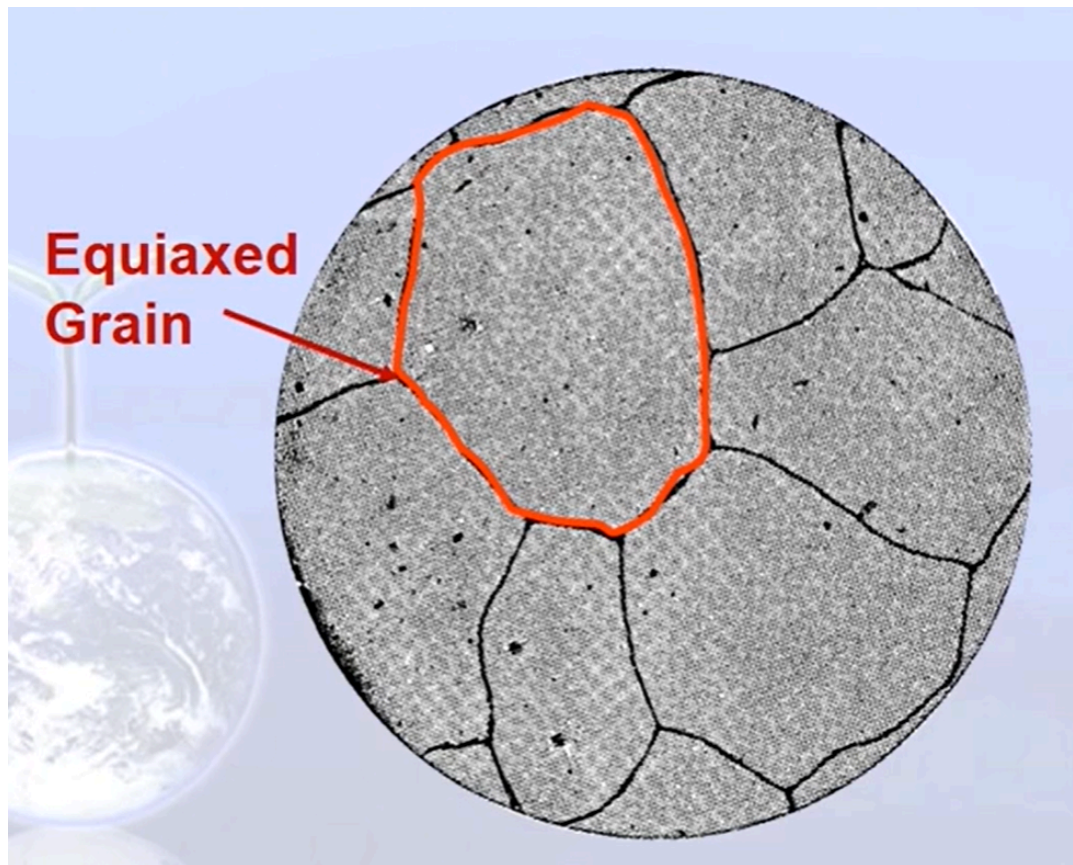
This causes more and more branches to form, making the crystal grow faster, causing the dendritic microstructure.

[https://www.doitpoms.ac.uk/tlplib/solidification\\_alloys/dendritic.php](https://www.doitpoms.ac.uk/tlplib/solidification_alloys/dendritic.php)

As these dendrites grow, they fill in more of the space between them, forming grains. Crystallization is complete. The dendrite structure is no longer visible from the grains.



## Grain Structure



The final cooled metal will have grains, consisting of a crystalline structure through the grain, but not throughout the metal. For this diagram, we call these grains equiaxed(equal axes), referring to approximately equal dimensions in all directions

The boundaries around the grains are called grain boundaries.

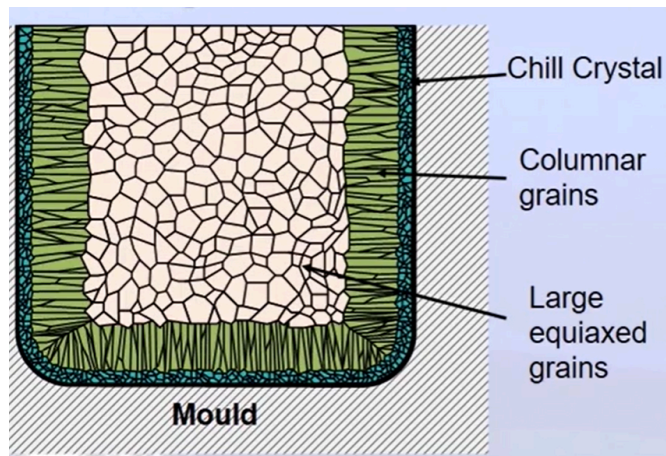
These grain structures are influenced by the processes used to make them, which causes different properties in the material due to the grain structure. This causes different performances in the part which the material is made out of

AKA:

Processes affect Structure which affects Properties which affects Performance

There are two processes that affect structure:

Cooling & Cooling Rates



When a metal is cooled in a mould, 3 types of crystals form.

The molten metal near the mould walls are cooled the first and the fastest, forming a thin film of metal at the mould walls.

The crystals in the metal film are called chill crystals. They are fine, equiaxed and randomly orientated.

More crystals then grow directionally from the cooler chill crystals to the core, forming columnar grains.

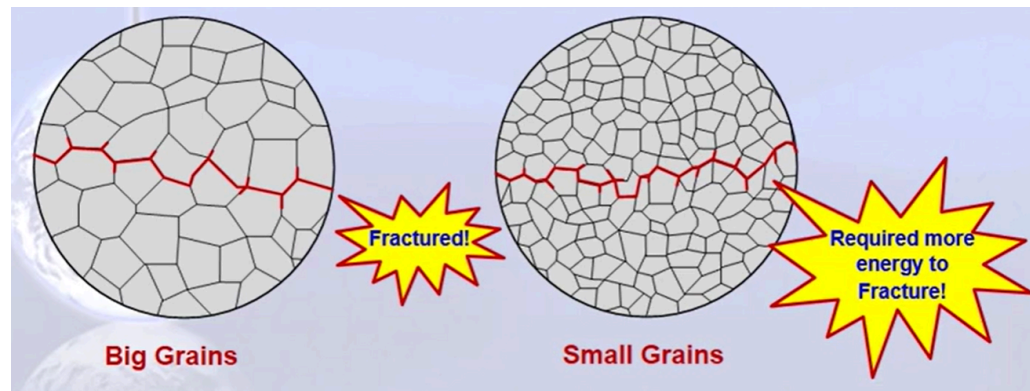
Then, in the core of the molten metal, where the cooling rate is slower, forms larger and coarser equiaxed grains.

Comparing larger and smaller grains, smaller grains will be stronger and tougher, with reasonable ductility.

This is because fractures can propagate more easily along the grain boundary of bigger grains than, due to fractures having to make less "turns" along the bigger grains, making it easier to propagate along the material.

This gives finer grain metals their strength, as they fracture less easily.

Finer grains also means that the metal can absorb more energy before it fractures, as the fracture needs more energy to propagate through, making finer grain metals tougher.

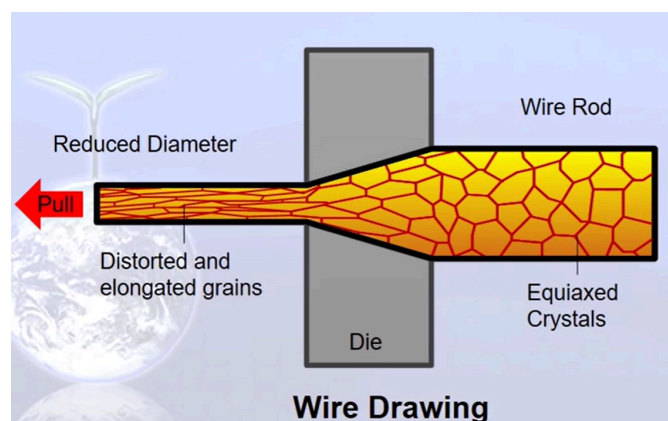


In summary: finer grain metals are tougher & stronger than coarse grain metals.

### Cold Working

Recall cold working: plastically deforming the metal below its recrystallisation temperature

During cold working, like wire drawing, the grains are originally equiaxed, however when pulled through a die, the grains deform when passed through the die and become distorted and elongated. The more the metal is deformed, the higher the degree of distortion and elongation of the grains.



Other cold working processes like rolling also results in work hardening, meaning that the final material is now harder and stronger, but brittle and less ductile.

### Plain Carbon Steels

Plain carbon steels are metal alloys of iron, containing other alloying elements like carbon, silicon and manganese

Other substances like sulfur and phosphorus (SP) may be found in pure carbon steels, however these are impurities, which are harmful.

The main factor that influences mechanical properties is the amount of carbon in the alloy.

some terms for this topic:

#### Solid/Liquid Solutions

Recall that liquid solutions are one or more substances dissolved in another to get a homogenous solution.

Thus, for solid solution, the same applies:

when the alloying element replaces the host atom(the main metal) such that it does not change the crystal structure, a solid solution is formed with the alloying element and the host atom.

#### Phase

Phase refers to any substance with a homogenous structure, regardless of state.

For example, a mixture of oil and water is not considered one phase, but two phases of oil and water, however salt and water is considered as one phase.

#### Allotropes

The ability for a substances to exist in more than one strucutre/physical form, which gives the substance different mechanical properties.

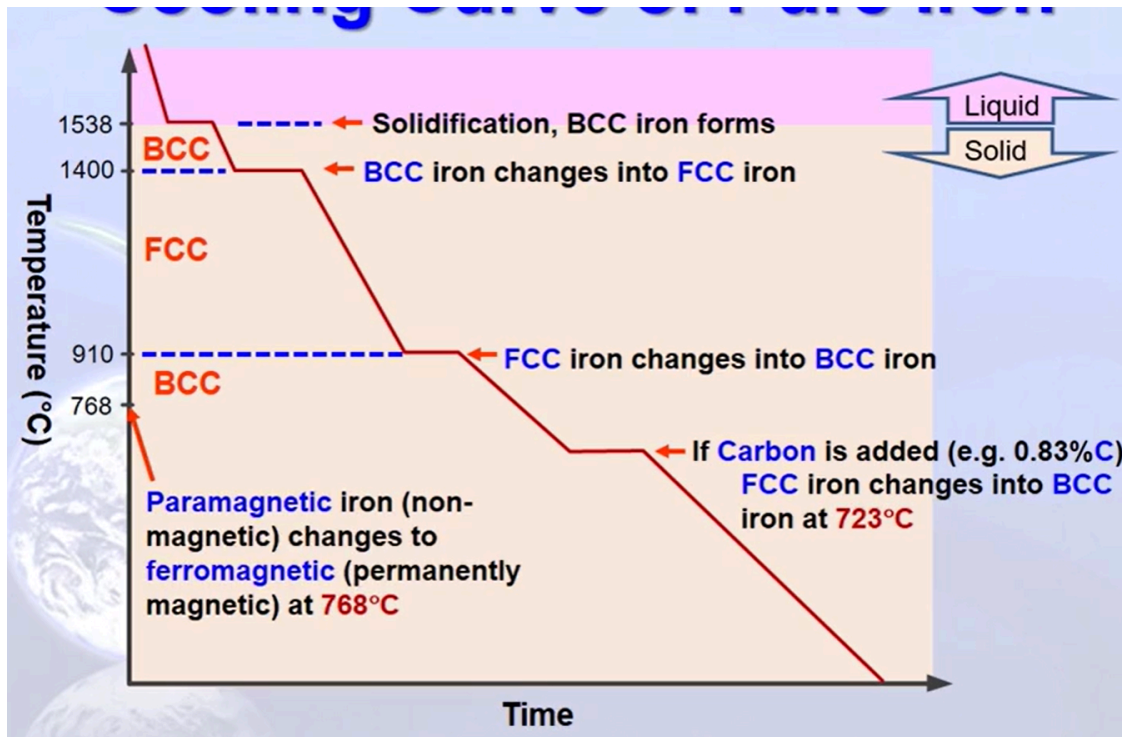
Examples are iron(different crystal structures), carbon(graphite and diamond) and titanium.

#### Phase Diagrams

##### Cooling curve of pure iron

As iron solidifies, its crystal structure changes at different tempeatures.





When iron solidifies at 1538C, it forms into BCC iron. Afterwards, its structure alternates between BCC and FCC

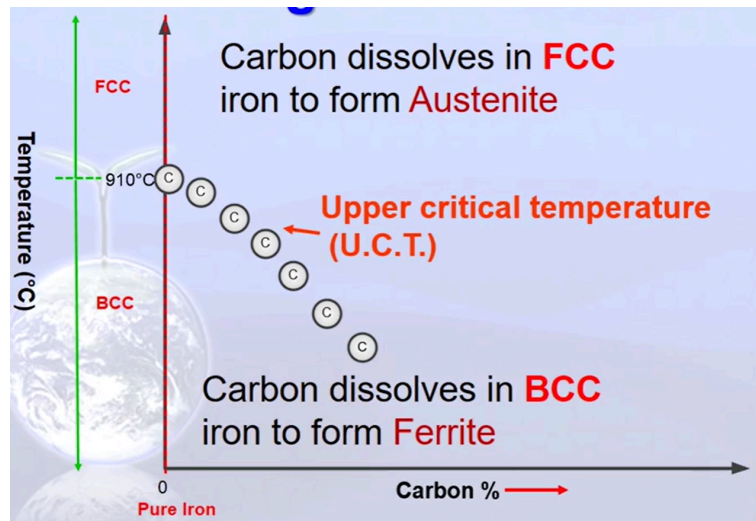
It first alternates and 1400C then at 910C.

However, this only applies for pure iron. If the iron consists of 0.83% Carbon, it does not alternate at 910C but instead lower at 723C. The more carbon there is, the lower the temperature is to alternate between FCC and BCC.

One thing to note is that above 768C, iron is paramagnetic, aka non magnetic, however below 768C iron is ferromagnetic, aka magnetic

Phase diagram: iron carbon:

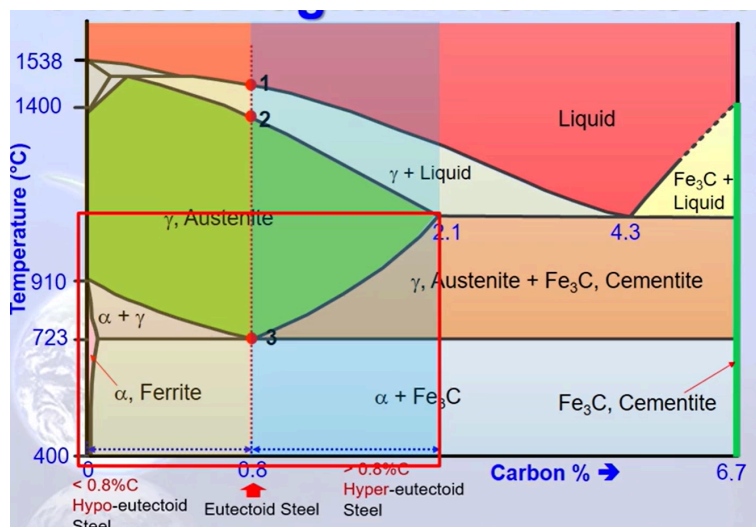




This section of the diagram illustrates how as more carbon is added, the transformation temperature between FCC to BCC lowers. This border is called the upper critical temperature (UCT).

Above the UCT, carbon dissolves in FCC iron to form austenite, while below the UCT carbon dissolves into BCC iron to form ferrite.

The full phase diagram of iron carbon is much more complex and shows more of the different types of structure the metal goes through depending on the temp and carbon amount.



Main things to note about this diagram:

The vertical line at 0 represents pure iron, while the vertical line at 6.7 represents cementite.

Cooling is represented as moving down a vertical line, from above 1538°C to 400°C. For example, at 0% C, the vertical line cuts through 3 boundaries, at

1538, 1400 and 910C, showing that the metal changes structures 3 times, excluding liquid form.

For steel with 0.8%C, it is called Eutectoid Steel.

From 0.022%C to 0.8%C, we call it Hypo-eutectoid steel (like hypothermia, low-temperature).

From 0.8%C to 2%C , we call it hyper eutectoid steel.

The red box shown will be the main focus of the phase diagram.

### Phases & their properties

#### Austenite( $\gamma$ -iron)

Solid solution of carbon in FCC iron

Austenite is soft and ductile, but nonmagnetic.

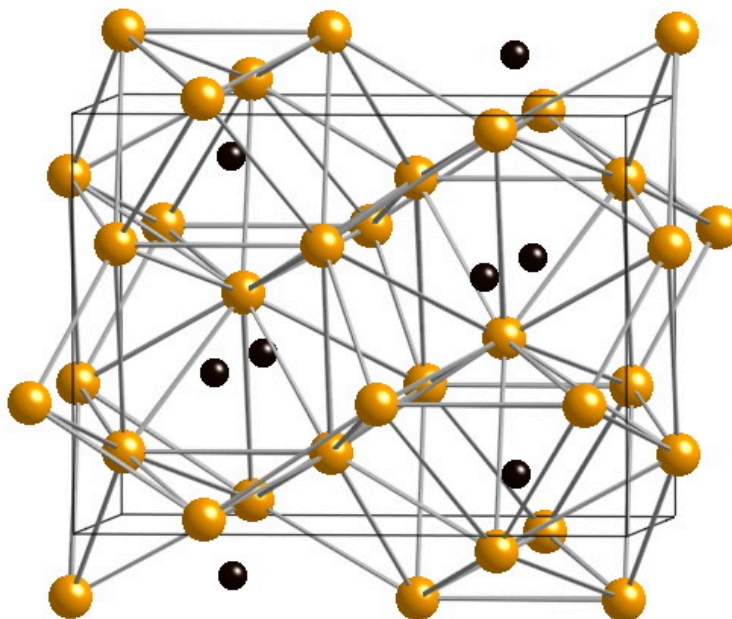
#### Ferrite ( $\alpha$ -iron)

Solid solution of carbon in BCC iron.

Also soft and ductile, however magnetic.

#### Cementite

A compound, Iron carbide ( $Fe_3C$ ), of a different structure called orthorhombic.

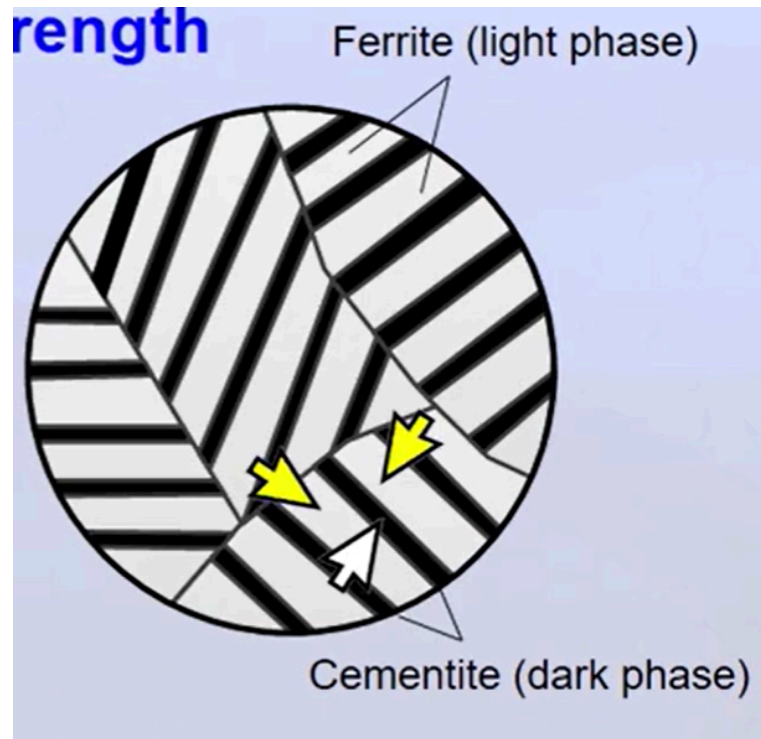


Cementite is very hard and brittle, opposite to austenite and ferrite

### Pearlite

Pearlite consists of a mixture of alternate layers (called alternate lamination) of ferrite and cementite.

Due to this structure, pearlite has a very high tensile strength



Pearlite is NOT a phase as it has two layers, however it is a microstructure

Summarising the properties of these steels:

|           | UTS | %E  | HBN |
|-----------|-----|-----|-----|
| Ferrite   | 340 | 40  | 100 |
| Cementite | 45  | NIL | 650 |
| Pearlite  | 930 | 5   | 280 |

From this, we can see that:

Ferrite is the most ductile, (high %E)

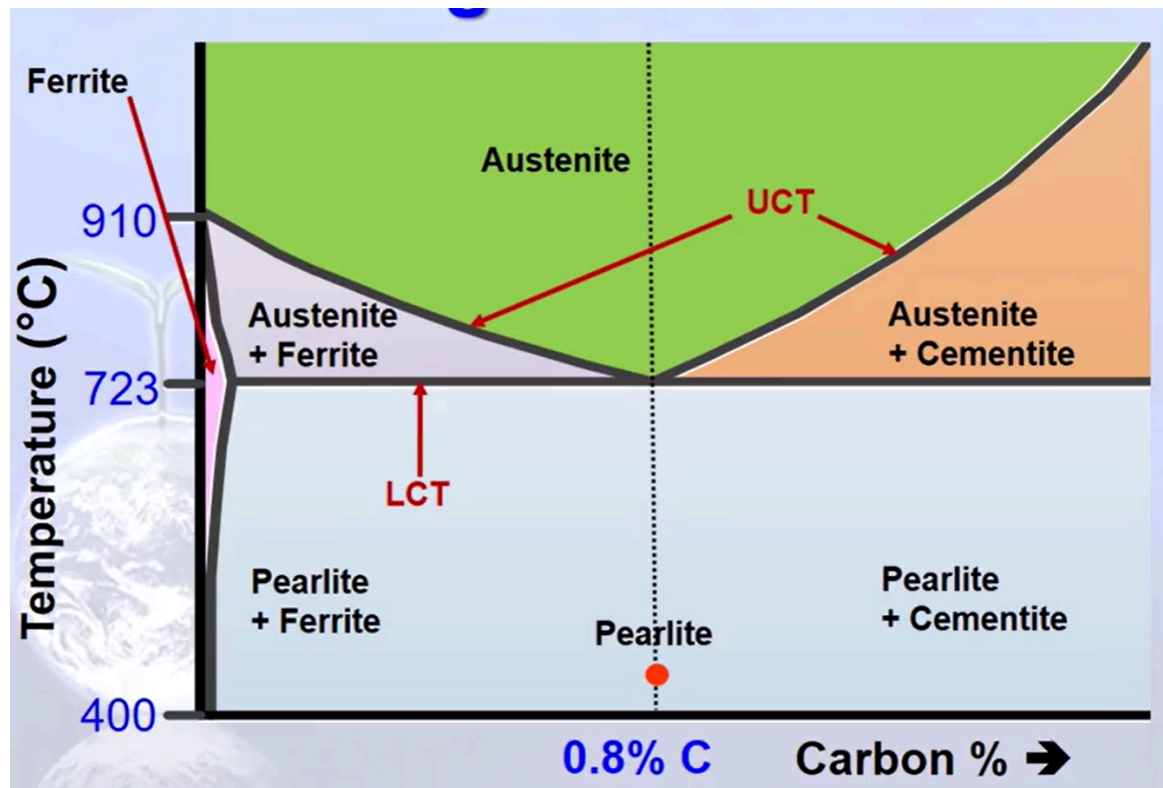
cementite is the hardest (high HBN)

pearlite is the strongest (high UTS)\

### Microstructures

The grain structure of eutectoid steels, hypo-eutectoid and hyper-eutectoid steels differs as they are solidified.

The highlighted section of the phase diagram shows how the structure of the steels changes as they are cooled:



### Eutectoid

Consider the vertical line at 0.8%C

As eutectoid steel cools, above the eutectoid point (723°C), the steel originally consists of austenite.

At the eutectoid point, austenite changes to pearlite. No other changes occur. Thus, eutectoid steel at room temperature is 100% pearlite.

This eutectoid point is also known as LCT, lower critical point.

Note that austenite changing to pearlite is a common thing for the other steels.

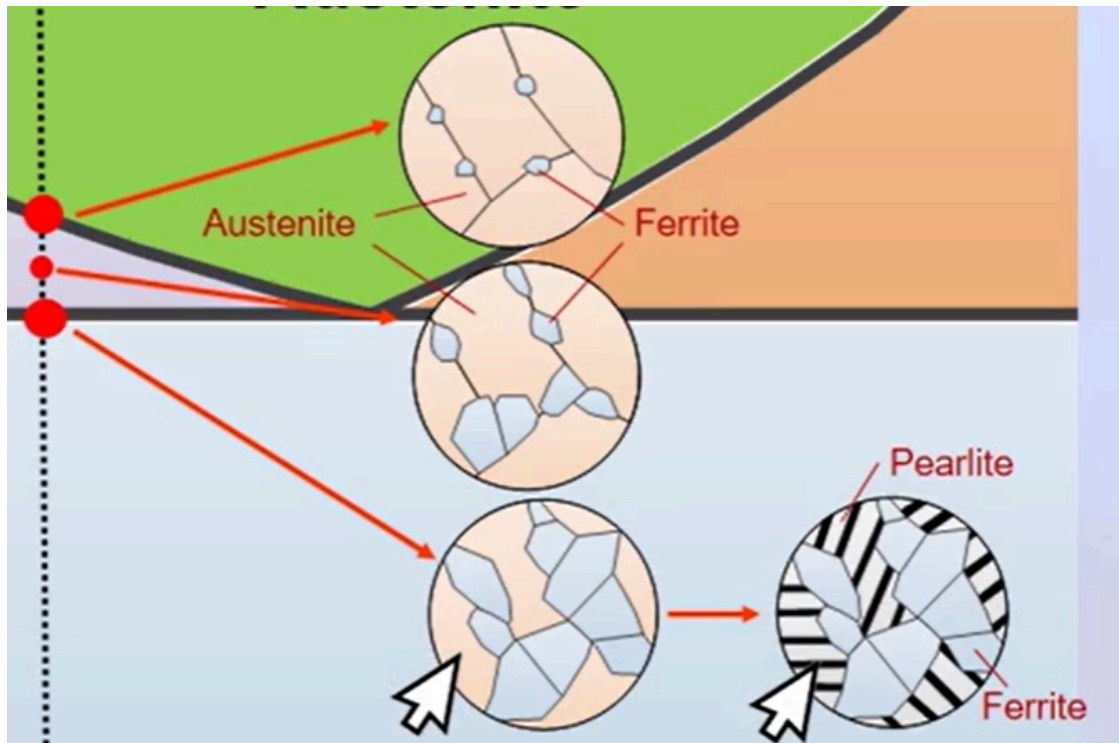
### Hypo-eutectoid

A vertical line at  $<0.8\%C$  will pass through the austenite and ferrite region, as well as the UCT and LCT.

This means that as the temp goes below the UCT, ferrite forms as it cools. As the temp drops, more ferrite forms until there are grains of both ferrite

and austenite.

below the LCT, ONLY the austenite changes to pearlite, leaving a mixture of pearlite and ferrite grains.

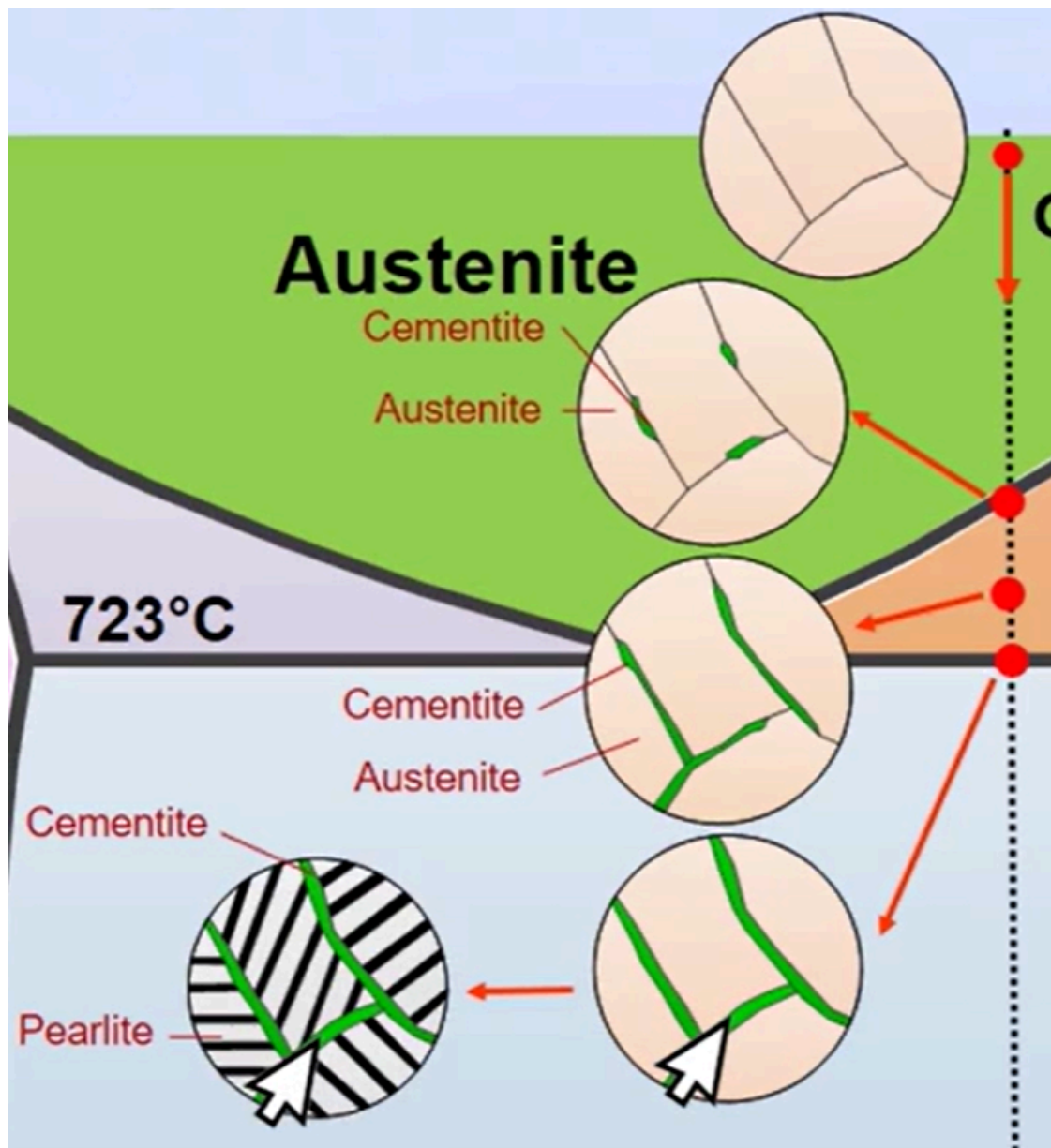


### Hyper-eutectoid

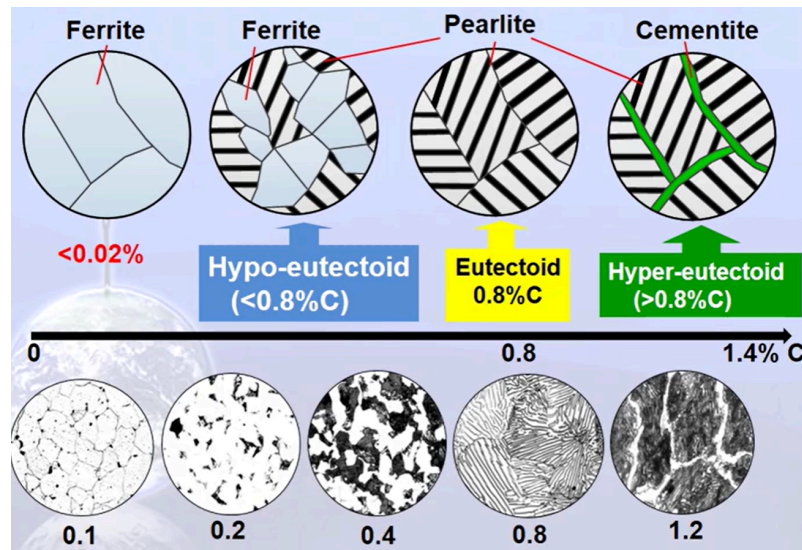
Similarly, a vertical line  $>0.8\%C$  passes through the cementite and austenite region, thus as the temp drops below the UCT, cementite forms along the grain boundary, and grow along them as the temp further drops.

Below the LCT, austenite turns into pearlite, leaving grains of pearlite and cementite.



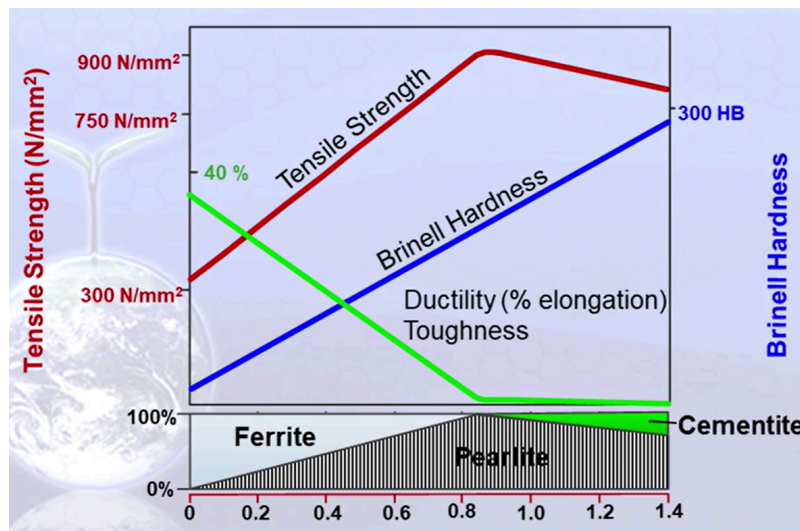


The diagram summarises the grain structure of the different steels.



### Mechanical properties due to carbon content

There are 3 main properties (tensile strength, hardness, and ductility) to take note of, and all other properties can be derived from others.



### Tensile strength

Tensile strength increases from 0 to 0.8C, then decreases afterwards.

Tensile strength is mostly dependent on the amount of pearlite, as pearlite has the highest UTS, and thus will affect the tensile strength of the steel the most. Thus, as %pearlite increases, strength increases and vice versa.

The increase of tensile strength is also due to the decrease in soft ferrite and the decrease in tensile strength at the end is due to the increase of brittle cementite.

### hardness

Hardness increases even past 0.8%C, as %pearlite increases until 0.8%C, then harder cementite increases afterwards.

### Ductility

Ductility decreases until 0.8C, where it stays constant.

As there is less ductile and soft ferrite as %C increases, the steel also becomes less ductile until 0.8C.

past 0.8C, there is no more ductile phases added/removed to the steel, so ductility stays the same.

For other mechanical properties, as %C increases

Weldability decreases

Machinability decreases (due to increased hardness)

corrosion resistance decreases

### Classifications & applications

all memo based, mostly due to convention:

| Carbon%   | Name                  | Properties                       | Uses                                           |                                                          |
|-----------|-----------------------|----------------------------------|------------------------------------------------|----------------------------------------------------------|
| 0.05-0.15 | Dead low carbon steel | soft and ductile                 | car bodies, rods, tubes, chains                | For things with complex shapes                           |
| 0.15-0.30 | low carbon steel      | soft and ductile                 | ship plates, reinforcing bars in buildings     | For things with complex shapes, however needing strength |
| 0.30-0.60 | medium carbon steel   | good mix of toughness & hardness | gears, crankshafts                             | For things under repetitive oscillations/ high fatigue   |
| 0.60-0.80 | high carbon steel     | high strength                    | railway rails, wrenches, screwdrivers, hammers | for parts with strength needed                           |
| 0.80-1.40 | carbon tool steel     | high hardness & wear resistance  | chisels, drills, razors, files, axes.          | for cutters                                              |